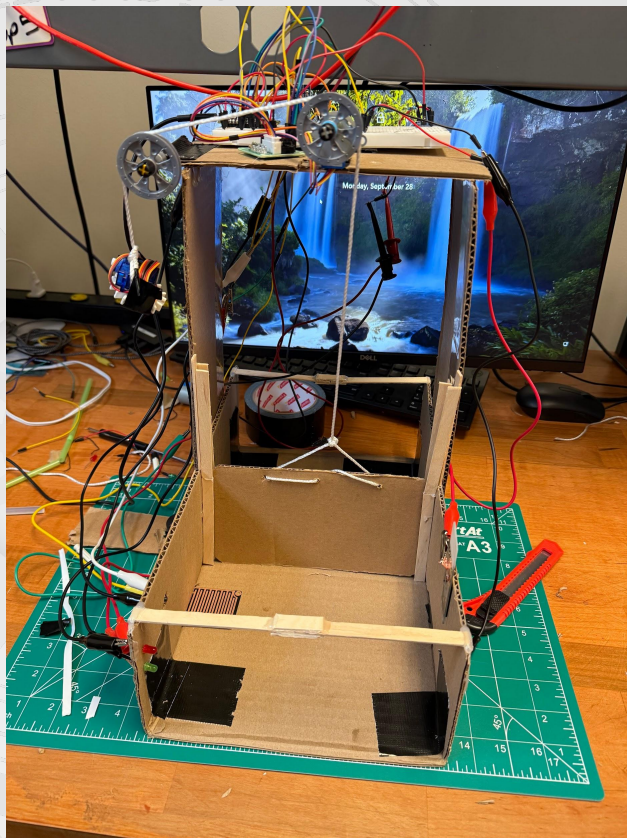
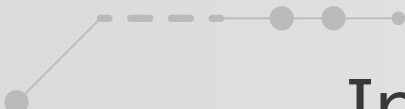




FYRE Project: Flood Gate

Diego Chuquillanqui, Artur Kis, Diane Ngo



Introduction & Project Overview

For our final project, we decided to make a **flood gate** that detects rising water and humidity, and automatically shuts.

Inputs:

- Rain sensor
- PEDOT humidity sensor
- Push button



Outputs:

- Stepper motor
- Status LEDs

Constraints:

- Structure Size
- Power delivery

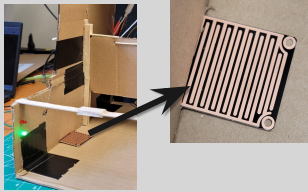
Requirements:

- Automated closure
- Visual Indication
- Smooth Actuation

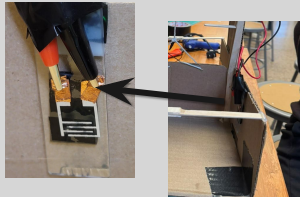


Implementation Overview

Sensing

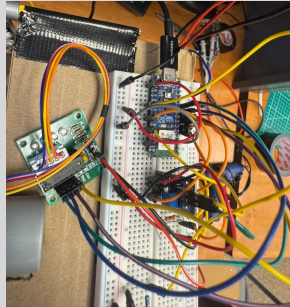


Rain sensor: Detects direct water contact



PEDOT:PSS sensor:
Detects humidity / moisture changes

Processing



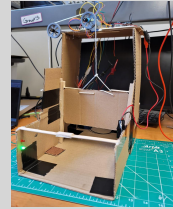
Microcontroller and circuit assembly

Reads both sensor inputs

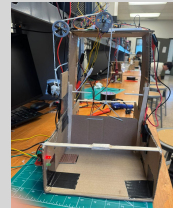
Checks against thresholds

Determines safe vs. unsafe conditions

Actuation

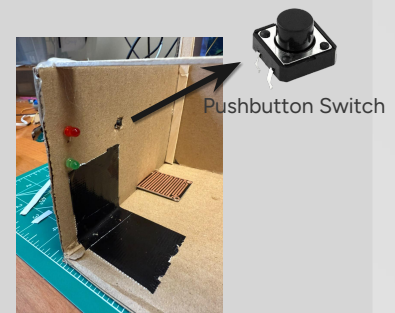


Safe: green LED on,
gate is open



Unsafe: red LED on,
motor closes the gate

Reset



Manual reset required
to reopen

Only works when
water is no longer
detected

Use of AI

Code Development

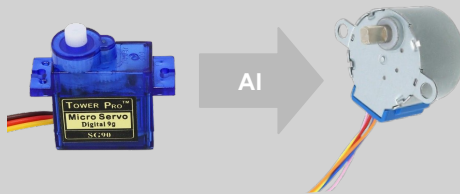
Generated and revised the full system code

Created test programs for sensors and motor behavior

- 01_stepper_motor_test.py
- 02_rain_sensor_test.py
- 03_pedot_sensor_response_test.py

Troubleshooting

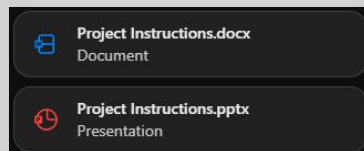
Identified and resolved motor selection, wiring, and driver issues



Proper AI Use

Provided relevant context: course materials, code, pins, etc

Reviewed suggestions before applying them



Why we used AI?

To focus primarily on:

 Sensor integration

 Testing

 Physical prototyping

 Improving the design



Results

Water Detection

- Wetting both the Rain Sensor and the PEDOT sensor reliability triggered the stepper motor to lower the gate.

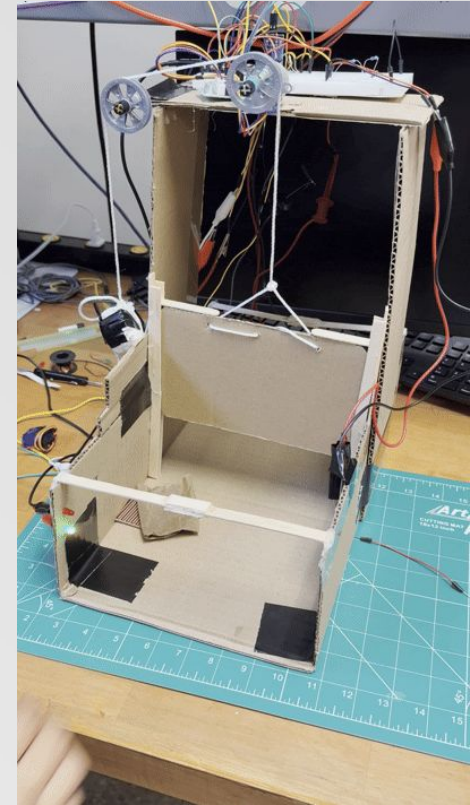
Manual Opening

- Pressing the button successfully toggled gate state when no moisture is detected → preventing accidental opening

Visual Feedback

- LEDs immediately switched states during gate trigger events.

Great Success!



Conclusion



Overall, our project ended up doing what we intended. What did we get out of it?

Humanitarian Impact:

- Successfully developed a functional, low-cost proof-of-concept flood gate that provides automated community protection without requiring manual on-site operation during flash floods.

Key Engineering Lessons:

- Realized the importance of guide alignment and pulley tension when lifting using stepper motors.

Future Improvements:

- Use 3D-printed plastic or acrylic instead of cardboard for full waterproofing and structural stability.
- Clean up wiring and integrate our own power supply.



Thank you!

Any questions, comments,
concerns, criticism?

